

Name: _____

Class: _____

Date: _____

Universal STEM Worksheet Engine

Secondary STEM Category | ISNSW Curriculum Conference 2026

Primary → Stage 6 · Pure TikZ, zero embedded images · NESAs-aligned



Scan or click:
AI Agent Competition

One engine, one design language — from a Year 2 child measuring a friendly stick-figure with a ruler, to a Stage 6 student resolving forces, analysing AC circuits and proving roots of unity. Every diagram is generated in **pure TikZ**: no embedded images. Outcome codes are **indicative drafts for teacher verification** against the current NSW NESA syllabuses. **Note:** Document layout is highly compressed for ISNSW showcase purposes. In-production artifacts generate with stage-appropriate spacing to manage cognitive load.

- Primary · K–6
- Junior STEM · Yr 7–10
- Senior STEM · Yr 11–12
- Extension

PRIMARY Early Years · Time & Measuring Length

Learning Intentions & Success Criteria

I can read analogue time, measure length in centimetres starting from 0, and compare heights using a scale.

Telling the Time

Y2

Analogue clock

[2 marks]

NESA: Measurement & Geometry

The clock shows a time in the morning. Write it in 12-hour and 24-hour formats.



12hr: _____ 24hr: _____

Using a Ruler

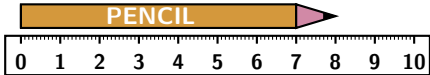
Y2

Measure pencil

[1 mark]

NESA: Measurement & Geometry

Read the ruler and state the exact length of the pencil in centimetres.



Length: _____ cm

Comparing Heights

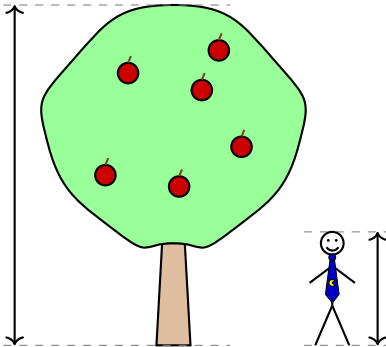
Y2

How tall?

[2 marks]

NESA: Measurement & Geometry

Use the scale to measure the **height** of the tree and Darra. How much taller is the tree?



Tree: _____ cm

Darra: _____ cm

Difference: _____ cm

JUNIOR STEM

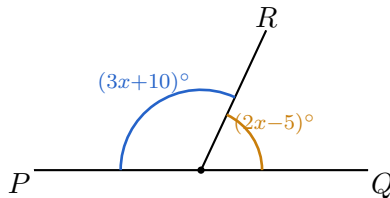
Stage 4–5 · Geometry, Algebra, Trigonometry & Science

Angle Reasoning

Stage 4 Angles on a line [2 marks]

NESA: Angle relationships

Line POQ is straight and ray OR meets it at O . Form an equation, solve for x , then state $\angle POR$.

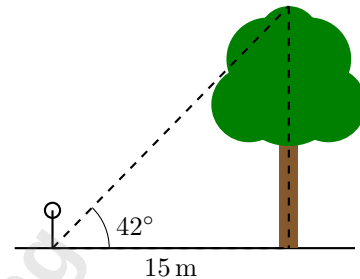


Angle of Elevation

Stage 5 Trigonometry [2 marks]

NESA: Trigonometry & Pythagoras

Standing 15 m from a tree, a clinometer reads 42° to the top. Find the height to 2 d.p.

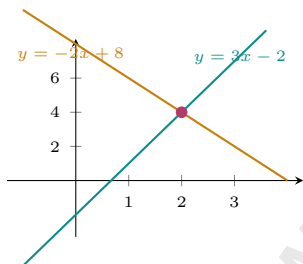


Simultaneous Lines

Stage 4 Point of intersection [2 marks]

NESA: Linear relationships

The lines $y = 3x - 2$ and $y = -2x + 8$ are graphed below. State their point of intersection.

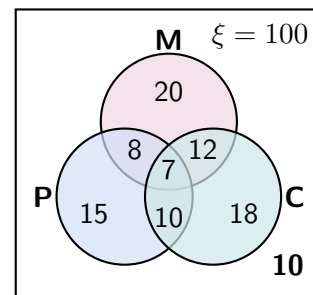


Set Theory & Probability

Stage 5 Venn diagrams [3 marks]

NESA: Probability / data

100 students study Maths (M), Physics (P) or Chemistry (C). Find $P(P | M)$.

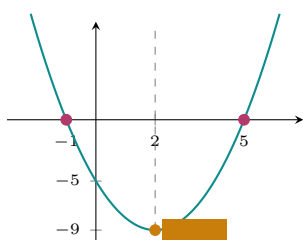


The Parabola

Stage 5 Quadratic features [3 marks]

NESA: Non-linear relationships

For $y = x^2 - 4x - 5$, state the x -intercepts, axis of symmetry, and the turning point.

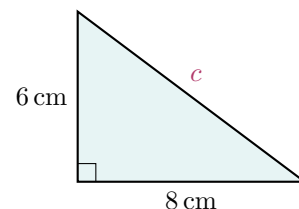


Pythagoras

Stage 5 Find the hypotenuse [2 marks]

NESA: Trigonometry & Pythagoras

A right-angled triangle has the two shorter sides shown. Find the length of the hypotenuse c .



$$c = \sqrt{a^2 + b^2}$$

Investigation Skills

Stage 5 Science — Pendulum Investigation

[4 marks]

A simple pendulum is released and you investigate how **string length** affects the **period** of one swing. State the independent and dependent variables, **two** controlled variables, a testable hypothesis, and **one** safety risk with its control. *NESA: Working Scientifically — questioning, predicting, planning*

IV: _____

Hypothesis / risk: _____

DV: _____

Controlled: _____, _____

SENIOR STEM

Stage 6 · Physics, Chemistry, Software & Extension

Learning Intentions & Success Criteria

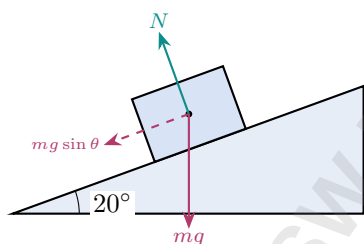
I can resolve forces on an incline, analyse a series RLC circuit at resonance, operate with 3D vectors, represent organic reactions, prove results about roots of unity, interpret a slope field, model a normal distribution, and reduce a logic circuit.

Mechanics · Inclined Plane

Physics Free-body diagram [3 marks]

NESA: Physics — Dynamics

A block of mass $m = 6.0 \text{ kg}$ rests on a *frictionless* incline at $\theta = 20^\circ$. Draw the free-body diagram and find the acceleration.

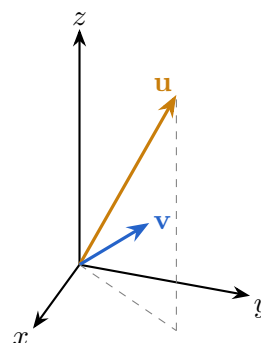


Extension · 3D Vectors

Maths Ext Dot product & angle [3 marks]

NESA: Enrichment — vectors in 3D

For $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, find $\mathbf{u} \cdot \mathbf{v}$ and the angle between them.

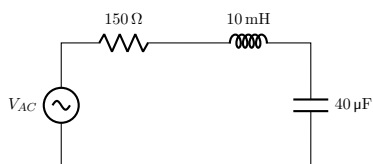


Electromagnetism · RLC

Physics Series resonance [3 marks]

NESA: Physics — Electromagnetism

For the series RLC circuit with $R = 150 \Omega$, $L = 10 \text{ mH}$ and $C = 40 \mu\text{F}$, calculate the resonant frequency in hertz.



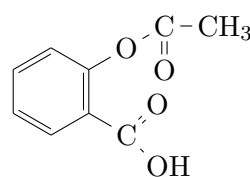
$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Chemistry · Organic

Chem Functional groups & synthesis [4 marks]

NESA: Chemistry — Organic Chemistry

Below is the skeletal structure of Aspirin. Identify the three primary functional groups present. Write the balanced chemical equation for the synthesis of Aspirin reacting salicylic acid with acetic anhydride.



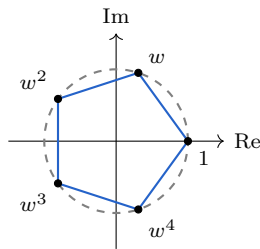
Equation: _____

Extension · Complex Numbers

Ext 2 Roots of unity [3 marks]

NESA: Mathematics Extension 2

The diagram shows the 5th roots of unity, where $w = \text{cis}\left(\frac{2\pi}{5}\right)$. Prove $1 + w + w^2 + w^3 + w^4 = 0$.

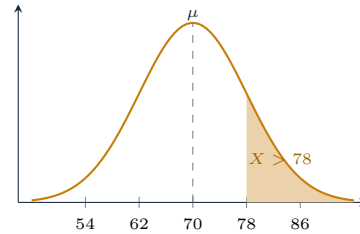


Statistics · Normal Distribution

Maths Adv The empirical rule [2 marks]

NESA: Mathematics Advanced — statistical analysis

Test scores are normally distributed with mean $\mu = 70$ and standard deviation $\sigma = 8$. Use the shaded region to estimate $P(X > 78)$.

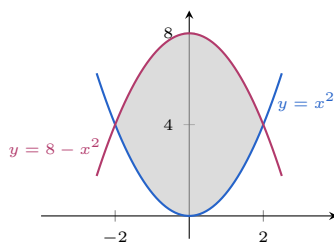


Calculus · Area Between Curves

Maths Adv Definite integral [3 marks]

NESA: Mathematics Advanced

Find the exact area of the shaded region bounded by $y = x^2$ and $y = 8 - x^2$.



Multiple Choice · Logarithms

Maths Adv Logarithmic equations [1 mark]

NESA: Mathematics Advanced

What is the solution to $\log_2(x) + \log_2(x - 3) = 2$?

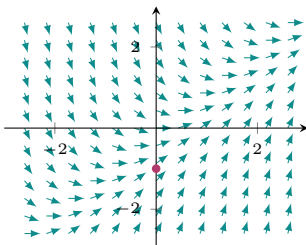
- A. $x = -1$
- B. $x = 4$
- C. $x = 4$ and $x = -1$
- D. No real solutions

Calculus · Slope Field

Maths Ext Direction field [3 marks]

NESA: Mathematics Extension 1 — differential equations

The slope field for $\frac{dy}{dx} = x - y$ is shown. Sketch the solution curve passing through $(0, -1)$ and describe its long-term behaviour.

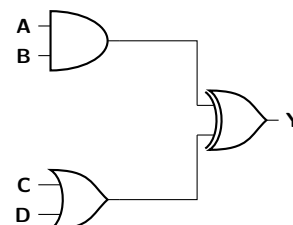


Software & Engineering

Soft Design Boolean circuit [3 marks]

NESA: Software Engineering

Write the Boolean expression for the output Y , then state the value of Y when $A = 1, B = 1, C = 0, D = 0$.



SHOWCASE Scalable Visual Pedagogy & Cross-Curricular**Learning Intentions & Success Criteria**

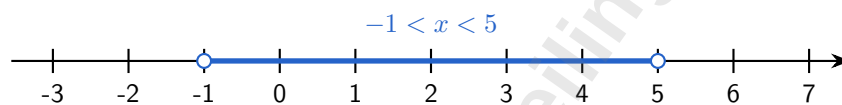
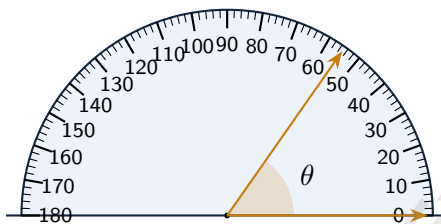
Competition Note: The following components demonstrate the engine's zero-shot ability to scale a single visual framework across distinct educational stages (Year 2 vs. Stage 6) and adapt to entirely different faculties (History, English) using pure TikZ.

Component 1 · The Number Line — One Macro, Any Stage

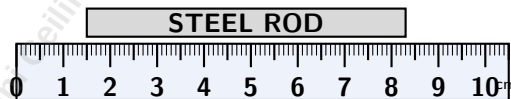
Year 2: Count the jumps to find where the kangaroo lands.



Stage 6: The *same* coordinate axis component now represents the solution of $|x - 2| < 3$.

**Component 2 · Visual Protractor**

Measure θ to the nearest degree.

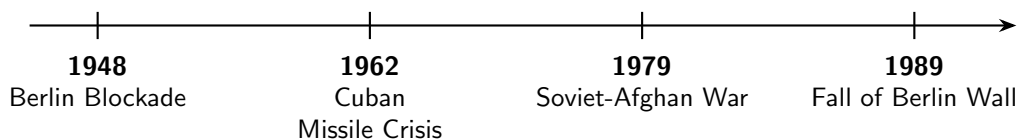
Component 3 · Precision Ruler

Read the rod's length to the nearest millimetre.

Cross-Curricular · Historical Timeline**Modern History** Timeline Analysis**[4 marks]**

NESA: Modern History

The timeline below maps four critical flashpoints of the Cold War. Select ONE of these events and evaluate its long-term impact on United States-Soviet Union geopolitical relations.



Cross-Curricular · Literary Analysis

English Adv Translation & Literary Devices**[4 marks]**

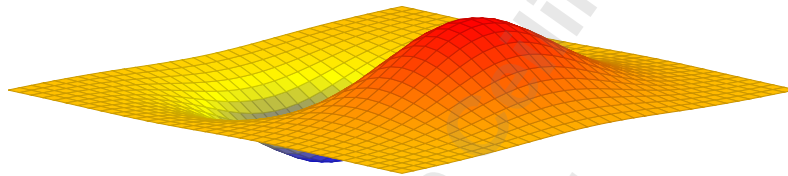
NESA: English Advanced

Translate the following excerpt from Shakespeare's *Hamlet* into modern Australian vernacular. Additionally, identify and analyse the primary literary device used in the final two lines to convey mental anguish.

William Shakespeare, Hamlet

To be, or not to be, that is the question:
Whether 'tis nobler in the mind to suffer
The slings and arrows of outrageous fortune,
Or to take arms against a sea of troubles...

Component 4 · Advanced 3D Topology & Data Vis



Advanced PGFPlots: Render complex 3D topological surfaces and multivariable calculus functions natively without relying on external image rendering engines.

Cross-Curricular · Business Studies

Business Strategic Management**[3 marks]**

NESA: Business Studies

A legacy automotive company has invested \$500 million into developing a highly efficient internal combustion engine. Before completion, a breakthrough in solid-state batteries makes electric vehicles significantly cheaper to produce. The board argues they must launch the new combustion engine to “recover our investment.” Identify the logical fallacy driving the board's argument and explain why proceeding could harm their long-term market position.

AI Pedagogy Engines · Deep Learning Extensions

Spot the Error Calculus bounds **[2 marks]**

A student evaluates $\int_0^2 2x\sqrt{x^2+1} dx$.

L1: Let $u = x^2 + 1$, $du = 2x dx$.

L2: $= \int_0^2 \sqrt{u} du$

L3: $= [\frac{2}{3}u^{3/2}]_0^2 = \frac{4\sqrt{2}}{3}$.

Error in Line: _____

Parameter Shift Dynamics **[2 marks]**

Refer to the **Inclined Plane** question. If the angle θ is increased to 30° and the mass doubled to 12.0 kg:

a) Does acceleration increase, decrease, or stay the same?

b) Does the normal force N increase or decrease?

ASSESS

Automated Marking System & Feedback

TikZ-Typeset Marking Rubric — Working Mathematically / Scientifically				
Criterion	Developing (1)	Proficient (2)	Exemplary (3)	Mark
Method	Partially valid approach	Selects a correct method	Efficient, fully justified method	
Working	Steps unclear or gaps	Logical steps, minor slips	Complete, clearly sequenced	
Accuracy & units	Frequent errors; units missing	Mostly accurate; correct units	Accurate; units & sig figs correct	
Communication	Hard to follow	Clear notation	Concise, syllabus-aligned notation	

Mark Allocation & Syllabus Map		
Band	Focus areas sampled	Marks
Primary (K–6)	Time, length, comparing height	5
Junior STEM (Yr 7–10)	Angles, linear, quadratic, trig, probability, Pythagoras, investigation	18
Senior STEM (Yr 11–12)	Dynamics, electromagnetism, organic chemistry, calculus, slope fields, statistics, logic	22
Extension	3D vectors, roots of unity	6
Total:		_____ / 51

What Went Well

Even Better If

Overall: ☐ Developing ☐ Proficient ☐ Exemplary

Grade: _____

AI Tutor Diagnostic · Cognitive Profile & Adaptive Remediation

Algorithmic Insight (Mental Maths Mentor):

Student demonstrates high procedural fluency in algebraic manipulation and calculus, but exhibits cognitive overload when translating text to visual-spatial representations (e.g., 3D Vectors, Geometry).

Adaptive Next Steps:

- Decrease: Repetitive procedural algebra tasks.
- Increase: GeoGebra-linked spatial reasoning tasks.
- Mentorship Note: High resilience shown in multi-step problem solving. Acknowledge effort and pivot to visual conceptual scaffolding for the next module.

Comm.

Algebra

Calculus

Logic

Spatial

ANSWERS

Teacher Answer Key — Worked Reasoning

Time. 12-hour: 10:10 AM; 24-hour: 10:10.

Ruler. 7.0 cm (the pencil tip aligns with the 7 mark; insist the student starts at 0).

Heights. Read each double-arrow scale: tree ≈ 6 units, Darra ≈ 2 units; the tree is about 4 units taller (accept consistent measured values).

Angles. POQ straight $\Rightarrow \angle POR + \angle ROQ = 180^\circ$: $(3x + 10) + (2x - 5) = 180 \Rightarrow 5x + 5 = 180 \Rightarrow x = 35$. Hence $\angle POR = 3(35) + 10 = 115^\circ$.

Lines. $3x - 2 = -2x + 8 \Rightarrow 5x = 10 \Rightarrow x = 2$, $y = 4$. Intersection (2, 4).

Parabola. $x^2 - 4x - 5 = (x - 5)(x + 1)$; intercepts $x = -1, 5$; axis $x = 2$; turning point (2, -9).

Elevation. $\tan(42^\circ) = \frac{h}{15} \Rightarrow h = 15 \tan(42^\circ) \approx 13.51$ m.

Venn. Only one subject: $20 + 15 + 18 = 53$. $P(P \mid M) = \frac{n(P \cap M)}{n(M)} = \frac{8 + 7}{20 + 8 + 12 + 7} = \frac{15}{47}$.

Pythagoras. $c = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$ cm.

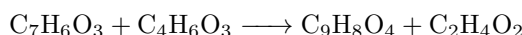
Pendulum. IV: string length; DV: period. Controlled (any two): bob mass, release angle, timing method, location. Hypothesis: longer string $\Rightarrow$ longer period. Risk: bob striking a student — keep the path clear, limit amplitude.

Inclined plane. Frictionless: $a = g \sin \theta = 9.8 \sin(20^\circ) \approx 3.35 \text{ m} \cdot \text{s}^{-2}$ down the plane ($N = mg \cos \theta$ balances the perpendicular component).

RLC. $f_r = \frac{1}{2\pi\sqrt{LC}}$, $L = 10 \times 10^{-3}$, $C = 40 \times 10^{-6}$: $\sqrt{LC} \approx 6.325 \times 10^{-4}$, so $f_r \approx 251.65$ Hz.

3D vectors. $\mathbf{u} \cdot \mathbf{v} = (2)(1) + (3)(2) + (6)(2) = 20$; $|\mathbf{u}| = 7$, $|\mathbf{v}| = 3$, so $\cos \theta = \frac{20}{21} \Rightarrow \theta \approx 18^\circ$.

Chemistry. Functional groups: carboxylic acid, ester, aromatic (benzene) ring. Synthesis:



(salicylic acid + acetic anhydride $\rightarrow$ aspirin + acetic acid).

Roots of unity. The roots solve $z^5 - 1 = 0$. Their sum is the geometric series $\frac{1 - w^5}{1 - w}$; since $w^5 = 1$ the numerator is 0 (and $w \neq 1$), so the sum is 0.

Area between curves. Intersect: $x^2 = 8 - x^2 \Rightarrow x = \pm 2$. $A = \int_{-2}^2 [(8 - x^2) - x^2] dx = \int_{-2}^2 (8 - 2x^2) dx = \frac{64}{3}$ square units.

Slope field. Solution through (0, -1) rises, crosses below the line $y = x - 1$, and is asymptotic to $y = x - 1$ as $x \rightarrow \infty$ (the particular solution is $y = x - 1$).

Normal distribution. $z = \frac{78 - 70}{8} = 1$. By the empirical rule $P(X > 78) = P(Z > 1) \approx 0.16$ (16%).

Logic. $Y = (A \wedge B) \oplus (C \vee D)$. With $A = B = 1$, $C = D = 0$: $(1) \oplus (0) = 1$.

MC Logarithms. B ($x = 4$). $x = -1$ is outside the log domain.

Spot the Error. Line 2. The integral bounds were not changed to u -values (should be $u = 1$ to 5).

Parameter Shift. a) Increases ($a = g \sin \theta$ is mass-independent, and $\sin(30^\circ) > \sin(20^\circ)$). b) Increases (doubling the mass outweighs the slight drop in $\cos \theta$).

Business Studies. Identifies the Sunk Cost Fallacy. Proceeding ignores future viability; EVs will undercut ICE vehicles, losing market share and compounding financial losses.



ISNSW AI Agent Competition



Try the Custom Gem!



Scan to view Agent Instructions